HW3

Ryan Thornburg

1a. 101110011

010001100 = -281

1b. 111111110 =

000000010 = -2

1c. 011101110 = 478

2. $x \oplus (y \oplus z) = (x \oplus y) \oplus z$

$x \oplus (y\bar{z} + \bar{y}z) = (\bar{x}y + x\bar{y}) \oplus z$

$\bar{x}(y\bar{z} + \bar{y}z) + x\overline{(y\bar{z} + \bar{y}z)} = \bar{z}(\bar{x}y + x\bar{y}) + z\overline{(\bar{x}y + x\bar{y})}$

$\bar{x}y\bar{z} + \bar{x}\bar{y}z + x((\bar{y}+z)(y+\bar{z})) = \bar{z}\bar{x}y + \bar{z}x\bar{y} + z((x+\bar{y})(\bar{x}+y))$

$\bar{x}y\bar{z} + \bar{x}\bar{y}z + x(zy + \bar{y}\bar{z} + F + F) = \bar{z}\bar{x}y + \bar{z}x\bar{y} + z(\bar{y}\bar{x} + xy + F + F)$

$\bar{x}y\bar{z} + \bar{x}\bar{y}z + xyz + x\bar{y}\bar{z} + F = \bar{x}y\bar{z} + x\bar{y}\bar{z} + \bar{x}\bar{y}z + xyz + F$

$$\boxed{\bar{x}y\bar{z} + \bar{x}\bar{y}z + x\bar{y}\bar{z} + xyz = \bar{x}y\bar{z} + \bar{x}\bar{y}z + x\bar{y}\bar{z} + xyz}$$

3.

00110110	54	01110101	117	11011111	-33
+ 01000101	69	+ 11011110	-34	+ 10111000	-72
01111011	123	01010011	83	10010111	-105